

NAME: Haseeb Chattha

SAP ID: 70136897

SUBJECT : Software Engineering

SUBMITTED TO :Sir Ali Ilyas

Agile Model:

The Agile Method is an approach to project management and software development that emphasizes iterative progress, adaptability, and collaboration with customers. Rather than completing the entire project after a lengthy timeline, Agile divides the project into smaller segments known as iterations or sprints, which usually last between 1 to 4 weeks. Each sprint results in a functional feature or increment of the product.

Structure:

1. Product Backlog:

A prioritized inventory of all features, requirements, and tasks necessary for the project, overseen by the Product Owner.

2. Sprint Planning:

The team selects items from the backlog to focus on in the upcoming sprint, establishes a sprint goal, and outlines how to accomplish the chosen tasks.

3. Sprints:

Time-boxed cycles (lasting 1-4 weeks) during which the team works on a specific set of tasks aimed at delivering a functional feature.

4. Daily Stand-ups:

Brief meetings (typically 15 minutes) where the team shares updates on progress, discusses upcoming tasks, and identifies any obstacles.

5. Sprint Review:

At the conclusion of each sprint, the team showcases the completed work to stakeholders and collects feedback.

6. Sprint Retrospective:

The team evaluates the sprint, discussing successes, challenges, and opportunities for improvement in future sprints.

7. Continuous Delivery:

Agile encourages frequent releases of a working product to facilitate ongoing user feedback and enhancements.

Advantages Of Agile Model:

1. Customer Satisfaction:

Agile facilitates the regular delivery of functional software or features, enabling customers to provide ongoing feedback.

Example: A customer requests a specific dashboard feature for a software product. In the first sprint, they receive a basic version of the dashboard and offer feedback for enhancements, which are implemented in the subsequent sprint.

2. Flexibility:

Agile accommodates changes in requirements, even late in the development cycle. This adaptability is essential for projects where customer needs or market dynamics may shift.

Example: Midway through a website project, the client asks for additional e-commerce features. Agile enables the team to integrate this new requirement without disrupting the overall project plan.

3. Risk Reduction:

By delivering features incrementally, potential issues or obstacles are identified early, preventing significant failures later on.

Example: During a sprint, the team realizes that integrating with a third-party API is more challenging than anticipated. They can quickly adjust their strategy without impacting the rest of the project.

4. Improved Collaboration:

Continuous communication between the team and stakeholders promotes strong collaboration and transparency.

Example: In an Agile marketing team, regular stand-up meetings enable designers, developers, and marketers to align and resolve issues in real time.

5. Faster Time to Market:

Agile supports the early delivery of product components, giving the company a competitive advantage by launching a working version sooner.

Example: A software company releases a Minimum Viable Product (MVP) after just a few sprints, allowing early adopters to start using it while development continues.

Disadvantages Of Agile Model:

Less Predictability:

Due to the evolving nature of Agile, it's hard to predict exact timelines and costs, especially for large, complex projects.

Example: A client wants to know when the software will be fully completed, but since new features keep being added during sprints, the team is unable to give a precise date.

High Commitment Required:

Agile requires continuous commitment from the team and stakeholders, including attending regular meetings and providing frequent feedback.

Example: In a small organization, the stakeholders do not have the time to provide regular feedback, which leads to misalignment between the team and client expectations.

Not Ideal for Small Projects:

Agile can be overkill for small, simple projects where iterative development isn't necessary.

Example: A freelance designer working on a one-page website may find that Agile's sprints and meetings add unnecessary complexity.

Documentation May Suffer:

Agile emphasizes working software over comprehensive documentation, which might lead to a lack of proper documentation if not carefully managed.

Example: A software team focuses on delivering features but doesn't document the architecture properly, causing issues when new developers join the project.

Dependency on Customer Availability:

Agile requires frequent customer input. If stakeholders are unavailable or provide unclear feedback, it can hinder progress.

Example: A product owner is frequently unavailable during sprint reviews, resulting in a delay in getting feedback on critical features.